

**BANARAS HINDU UNIVERSITY**

B.Sc. (Hons.) Semester V Examination, 2016-17

**Paper No. BPT-505: Electromagnetic Theory**

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. Symbols have their usual meanings.***Question 1.** Answer any **four** of the following:

[4 × 5 = 20]

- (a) Show that the vector field  $\vec{v} = yz\hat{i} + zx\hat{j} + xy\hat{k}$  can be written as the gradient of a scalar field  $f(x, y, z)$ .
- (b) If an infinitely long straight wire carries a uniform line charge density  $\lambda$ , find the electric field at a distance  $r$  from this wire.
- (c) If the potential due to a continuous charge distribution is written as  $V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|} d\tau'$ , obtain  $\nabla^2 V(\vec{r})$ .
- (d) A parallel polarized electromagnetic wave is incident from air to polystyrene ( $\mu \approx \mu_0$ ,  $\epsilon_r = 2.6$ ) at the Brewster angle. Determine the transmission angle.
- (e) Determine the state of polarization for the wave whose electric field is given by:

$$\vec{E}(z, t) = E_0(\hat{i} \cos(kz - \omega t) - \hat{j} \sin(kz - \omega t))$$

- (f) Calculate the intrinsic impedance of free space.

**Question 2.**

- (a) What is a linear dielectric medium? How do Maxwell's equations modify in such a linear dielectric medium? [6]
- (b) Does the Poynting vector correspond to a dual character in the energy-momentum conservation laws in electrodynamics? Explain in detail. [6]

**Question 3.**

- (a) In a lossless medium for which intrinsic impedance  $\eta = 60\pi$ , relative permeability  $\mu_r = 1$ , and magnetic field:

$$\vec{H} = -0.1 \cos(\omega t - z)\hat{i} + 0.5 \sin(\omega t - z)\hat{j} \text{ A/m}$$

find the relative permittivity  $\epsilon_r$  and the angular frequency  $\omega$ . [8]

- (b) What is a phasor? The electric and magnetic fields in free space are given by  $\vec{E} = E_0 \cos(10^8 t + \beta z)\hat{i}$  V/m and  $\vec{H} = H_0 \cos(10^8 t + \beta z)\hat{j}$  A/m. Find the phasor form of these fields. [6]

**Question 4.**

- (a) How do the laws of electrostatics enforce boundary conditions on the electric field and the corresponding electrostatic potential? [6]
- (b) Why does any net charge reside entirely on the surface of a conductor? Explain. [6]

**OR**

- (a) Using the general solution of the Laplace equation in spherical polar coordinates with azimuthal symmetry, determine the potential outside the surface of a hollow empty sphere of radius  $R$  if the surface potential is maintained at  $V_0(\theta) = V_0 \sin^2(\theta/2)$ , where  $V_0$  is a constant. [8]
- (b) For a uniform magnetic field  $\vec{B}$ , show that  $\vec{A} = -\frac{1}{2}(\vec{r} \times \vec{B})$  is a valid vector potential such that  $\nabla \cdot \vec{A} = 0$  and  $\nabla \times \vec{A} = \vec{B}$ . [6]
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**Question 5.**

- (a) A standard air-filled rectangular waveguide with dimensions  $a = 8.636$  cm and  $b = 4.318$  cm is fed by a 4 GHz carrier wave. Determine if the  $TM_{11}$  mode will propagate. If so, calculate the phase and group velocities of this mode. [8]
- (b) Show that a rectangular waveguide does not support  $TM_{10}$  and  $TM_{01}$  modes. [6]

**OR**

- (a) Derive the general expressions of the attenuation constant  $\alpha$  and phase constant  $\beta$  for electromagnetic wave propagation in conducting media. [8]
- (b) Prove the relation  $R + T = 1$  for normal incidence of electromagnetic waves at the boundary between two media. [6]
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